An Evaluation Artefact in Presence–Background Archaeological Modelling: Evidence from East Java and a Corrected Comparison Protocol

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Abstract

Changing how background points are selected can raise the cross-validated AUC of a presence–background model, but such an increase does not necessarily mean that the model has become better. This study tests how the background design and the evaluation background affect model evaluation, using 378 archaeological sites in East Java, three algorithms (XGBoost, RandomForest and Maximum Entropy) and spatial block cross-validation. In an initial sequence of seven experiments, the reported AUC rose overall from 0.659 to 0.751 (the latter a mean over seeds), and the rise was attributed to increasingly realistic background designs. When all designs are evaluated on a common background, changing the background design contributes −0.014 AUC, whereas adding distance to rivers as a feature contributes +0.042. The gain of the final, hybrid design is therefore an evaluation artefact, and the gains of the earlier target-group designs are not reproduced. In synthetic data with a known truth, the hybrid design evaluated on its own background is reported above its true performance in all 240 of its runs. The AUC on a design's own background, the criterion used to choose a design, reaches its maximum at the end

of the range tested, so it gives no indication of when to stop adding hard negatives (background points selected far from the typical site environment). Priority maps from a single random seed are unstable; averaging seeds helps, but two independent five-seed averages still agree at a Jaccard overlap of only 0.76–0.86 on the top 10% of cells. Above about 1,000 m, elevation and proximity to a Quaternary volcanic centre cannot be adequately separated in the study area. We propose a protocol: hold the evaluation background fixed when comparing designs, state the availability domain of every metric, and average priority maps over seeds. The contribution is this diagnosis and protocol, not a new prediction of site locations.

Keywords: presence-background modelling; pseudo-absence design; evaluation artefact; spatial cross-validation; survey bias; East Java

1 Introduction

Predictive settlement modelling in Java is difficult because the observed archaeological record is shaped as much by preservation and discovery as by past settlement behaviour. Known site maps are incomplete and strongly conditioned by survey history; active volcanism and tephra deposition can bury older surfaces; and from a computational standpoint the combination produces label noise in pseudo-absence design and inflates validation when spatial dependence is mishandled. Changing how background points are selected can raise the cross-validated AUC, but such an increase does not necessarily mean that the model has become better. The AUC can rise because the change in background design makes the evaluation task easier, not because the model has better predictive ability. This study aims to distinguish between these two possibilities by testing how the background design and the evaluation background affect the outcome of model evaluation, using presence–background modelling of the distribution of archaeological sites in East Java.

1.1 What presence–background models are, and why evaluating them is difficult

Presence–background (presence-only) models learn a suitability surface from known presences and a set of background points that stand in for the available environment. They are the workhorse of ecological niche modelling (ENM) and species distribution modelling (SDM) (Elith et al. 2006; Phillips, Anderson & Schapire 2006) and of much archaeological predictive modelling (Kvamme 2006; Verhagen & Whitley 2012). They are acutely sensitive to sample-selection bias, especially when the background does not reflect the same observation process as the presences (Araújo & Guisan 2006; Lobo, Jiménez-Valverde & Hortal 2010; Phillips et al. 2009; Wisz & Guisan 2009). The standard correction for survey-biased data is target-group background (TGB): draw background points from the area a survey could plausibly have reached, so that the model learns presence against a realistic 'available' set rather than against the whole landscape (Phillips et al. 2009).

A model improved in this way is judged by cross-validation against held-out points, and the held-out *negatives* are drawn from the same background distribution the model was trained on. This is where the delicacy lies. If two background designs are each scored against their own background, the comparison is not identified: a design that makes its own background easier to separate from presences will look better, for a reason that has nothing to do with how well either model captures the phenomenon. Random train/test splits overestimate transfer when observations are spatially clustered, because held-out points sit close to training points (Hijmans 2012; Roberts et al. 2017); spatially structured cross-validation is the recommended corrective (Valavi et al. 2019). But spatial blocking fixes where the folds are; it does not, by itself, fix *which* background the held-out negatives come from. That second choice is the subject of this paper. Table 1 defines the key terms used throughout the paper.

1.2 Four reasons a site may be absent, and which one we model

A site may be absent from the record for four distinct reasons, and conflating them is the central interpretive hazard of this literature:

Term	Definition
Presence–background model	A model trained on sites whose presence is known (presences) and on background points whose presence status is unknown, so that it produces a relative suitability score, not a probability that a site is present.
Background design	The rule for selecting background points: *random* samples uniformly across the whole study area; *TGB* weights the selection by distance to roads, with an exponential decay scale of 12 km; *hybrid* combines the TGB pool, a quota per region and hard negatives, with a target share of 30%.
Hard negatives	Background points whose environment lies at a standardised distance of 2–5 from the mean site environment (z, in standard-deviation units across the model's features; Section 2.3). In this paper the term denotes points selected by their distance from the typical site environment, not, as usually in machine learning, points that resemble the sites.
Evaluation background	The background points used to assess model performance on test data: an *own* background uses the same design as the one the model was trained with, whereas a *common* background uses one fixed set of points for all designs, in this study mainly a uniform sample.
Availability domain	The geographical area from which background or evaluation points are drawn; it sets the context of the comparison, so the value of an evaluation metric must be interpreted relative to that area (Barve et al. 2011).
AUC (area under the ROC curve)	The probability that a randomly chosen site receives a higher score than a randomly chosen background point; 0.5 is equivalent to random guessing and 1 indicates perfect separation. Its value depends on the background points used for evaluation.
TSS (true skill statistic)	A measure of classification performance, calculated as sensitivity plus specificity minus one at a given classification threshold.
Continuous Boyce index	A presence-only evaluation metric that compares the observed frequency of sites with the frequency expected across suitability classes, relative to a fixed uniform sample, so that its reference sample does not depend on the training background design.
Spatial block cross-validation (CV)	A cross-validation method that divides the study area into five folds made of spatial blocks of about 50 km (0.45°), to reduce the geographical proximity between sites in the training and the test data.
Seed ensemble	A combined map obtained by averaging the predictions of several random seeds. In this study, two ensembles of five seeds that share no seed agree at a Jaccard overlap of 0.76–0.86 on the cells in the top 10% of the ranking, against 0.55–0.70 for two single seeds.
Jaccard overlap	A measure of similarity between two maps, calculated by dividing the number of cells in the top 10% of both maps by the number of cells in the top 10% of at least one of them.
Coarsened exact matching	A matching method that compares two groups only within strata that contain members of both groups, with weights based on the size of the smaller group; in Section 3.9 it is applied within each 200 m elevation band.

Table 1: Key terms.

1. **Environmentally unsuitable** — the location was never attractive for settlement.

2. **Not surveyed** — a site could be there, but no one has looked there.

3. **Buried** — a site was there and is now below the detectable surface.

4. **Destroyed** — a site was there and has since been erased.

This paper models suitability only (mechanism 1). The other three are the interpretive frame, not the output of the model. A low suitability score does not distinguish 'never settled' from 'buried' or 'destroyed'; a high suitability score does not assert that a site exists.

1.3 Contribution and relation to prior work

Every background design is benchmarked against Maximum Entropy (MaxEnt), a widely used presence–background modelling method (Elith et al. 2006), on the same cross-validation folds; this comparison is the decisive analysis of the study. An initial analysis showed a 'ladder': a sequence of seven experiments (E007–E013) whose reported AUC rose from 0.659 (E007) to 0.751 (E013, mean over seeds), with a single dip at E009. However, the results in Section 3.2 show that the part of this rise attributed to increasingly realistic background designs is an evaluation artefact, not evidence that such designs produce better models. Thus the same empirical material now supports a more limited conclusion: the assessment of presence–background model performance depends strongly on the background used to evaluate it; *background designs in presence–background modelling cannot be compared unless the evaluation background is held fixed, and a selection criterion computed on a design's own background has no interior optimum.* The selection criterion, AUC computed on a design's own background, does not produce an optimum inside the range of hard-negative shares: in the synthetic worlds the AUC keeps rising from 0.7367 to 0.8904 as the share is raised from 0 to 1, whereas on the real data the AUC rises from a share of 0.1 onwards, with one small decrease of 0.0071 between shares of 0 and 0.1, so the maximum always lies at the end of the range tested and the criterion gives no indication of when to stop adding hard negatives.

Our contribution sits *inside* the ENM literature rather than beside it, and we state its boundary precisely, because this family of problems is well established in adjacent fields.

It is established. Lobo, Jiménez-Valverde & Real (2008) showed that AUC is not comparable between models whose background differs, because the metric is computed over whatever availability domain the modeller supplied; Barve et al. (2011) formalised that domain as the accessible area M and showed that its delineation conditions both the fitted model and its evaluation; and Hijmans (2012) demonstrated that the spatial arrangement of background relative to presences inflates cross-validated AUC, proposing a null-model calibration for it. Model-selection criteria computed within a single background are likewise known to be unreliable guides to transferability (Radosavljevic & Anderson 2014; Warren & Seifert 2011), and discrimination metrics in general are known to answer a narrower question than applications ask of them (Guillera-Arroita et al. 2015; Lobo, Jiménez-Valverde & Hortal 2010). That the target-group correction is standard (Phillips et al. 2009) and that pseudo-absence choice moves reported performance (Barbet-Massin et al. 2012; Senay, Worner & Ikeda 2013) are both settled points. The transfer of species-distribution modelling into archaeology is equally established, from eco-cultural niche modelling (Banks et al. 2006), through MaxEnt-based prediction of archaeological site distributions (Howey, Palace & McMichael 2016; Noviello et al. 2018; Yaworsky et al. 2020), to the standard methodological reference (Franklin 2009). Recent applied studies report fair to good discrimination when predicting site locations from environmental variables (AUC of about 0.7 to 0.9; Castiello & Tonini 2021; Wang, Shi & Oguchi 2023), and Comer et al. (2023) report that the way samples are drawn for training and testing can itself change the measured performance.

This study makes three contributions. First, we execute on one dataset the comparison this literature implies but rarely runs end to end: the same designs, the same folds, scored both on their own background and on a common one, so the artefact is *measured* rather than argued (Section 3.2). Second, we characterise the criterion a modeller actually optimises (AUC on the design's own background) and show it has *no interior optimum*, so it cannot tell them when to stop; we are not aware of a comparable measurement of

reported against true performance as the share of hard negatives in the background is varied (Section 3.3). Third, we give the effect a size in an archaeological setting: the inflation is of the same order as the gain that we, and others, report as a result, which is why this is a substantive hazard and not only a metrological caveat.

1.4 Study area

The study area is province-scale East Java (approximate bounds 111–115°E, 9–6.5°S), processed in EPSG:32749. East Java's archaeological record is dominated by the Hindu–Buddhist period (roughly the 8th–15th centuries CE), including the Singhasari and Majapahit kingdoms, whose temples and inscriptions concentrate in the Brantas river valley around Malang–Mojokerto–Blitar (e.g. Candi Penataran, Singhasari, Jago and Kidal). The known record is therefore both chronologically and spatially selective: it samples the well-surveyed lowlands more densely than either the volcanic highlands (Kelud, Semeru, Bromo–Tengger) that surround them or the karstic southern mountains, and discovery history is strongly conditioned by modern road access. The 378 usable presences in this study are the covariate-valid subset of that record; Figure 1 shows how unevenly they cover the province.

2 Materials and Methods

2.1 Data, covariates, and the principle of role separation

Usable presences after covariate filtering were 378 (111–115°E); the soil-coverage subset used once in E009 was 259. Every variable plays one analytical role, and the roles must not be mixed, because mixing them makes a suitability score uninterpretable: a low score could mean the place was unsuitable, or unsurveyed, or not preserved, and the model cannot say which. We therefore keep only the suitability block as predictors, express accessibility through the background design (Section 2.4), and treat preservation and volcanic proximity as diagnostics applied to the output. Table 2 assigns every variable a single role.

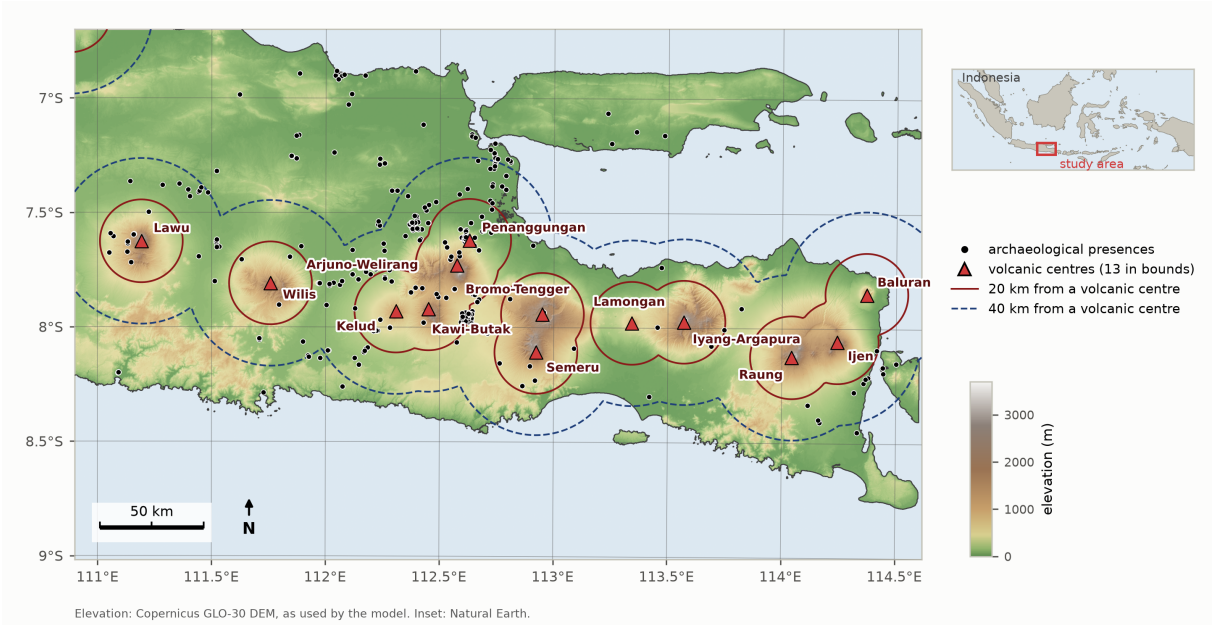


Figure 1: Study area in East Java, Indonesia, showing the presences used for modelling and the 13 canonical volcanic centres inside the paper's stated bounds (111–115°E), used as post-hoc diagnostics only.

This separation is not a technicality. It is central to this paper: the variable that defines the target-group background (accessibility, through `road_dist`) is kept out of the features, so accessibility can affect the reported numbers only through the background design, which is what Section 3 examines.

2.2 The modelling pipeline

We model binary presence/background suitability. The pseudo-absence (background) ratio is fixed at 1:5 throughout. Three learners are compared: XGBoost (Chen & Guestrin 2016) (primary), RandomForest (Breiman 2001), and a Maximum Entropy model (Elith et al. 2006; Phillips, Anderson & Schapire 2006). The Maximum Entropy model is the `elapid` implementation of maxnet, with linear, hinge and product features, regularisation multiplier $\beta = 1.5$ and a cloglog transform; β is swept from 0.5 to 4.0 in Section 3.2 to confirm the result is not an artefact of that setting. Validation uses 5-fold deterministic spatial block cross-validation at a baseline block size of 0.45° (∼50 km). Primary metrics are AUC and the True Skill Statistic (TSS), the latter preferred to kappa because it does not vary with prevalence in the way kappa does (Allouche, Tsoar & Kadmon 2006). Both

Variable	Source	Analytical role	How it enters
elevation	Copernicus GLO-30 DEM, 30 m, decimated ×10 (∼300 m)	Suitability (terrain)	training feature
slope	derived from DEM	Suitability (terrain cost)	training feature
TWI	topographic wetness index, derived from DEM	Suitability (water availability)	training feature
TRI	terrain ruggedness index, derived from DEM	Suitability (terrain cost)	training feature
aspect	derived from DEM	Suitability (insolation/orientation)	training feature
river_dist	OSM/HydroSHEDS, Euclidean distance	Suitability (water access); added at E008	training feature
clay, silt	SoilGrids topsoil	Preservation / substrate	features in **E009 only**; dropped (lowered AUC)
road_dist	OSM roads (expanded class set from E012)	Accessibility / survey effort	**never a training feature**
volcano_dist	canonical inventory, 30 centres (13 in bounds)	Taphonomy / burial — diagnostic only	**never a training feature**

Table 2: Covariates and their analytical roles. Roles are exclusive by construction: a suitability variable answers *would people have settled here?*; an accessibility variable answers *would we have found it?*; a preservation variable answers *would it still be there to find?*. TWI (topographic wetness index) quantifies the tendency of a cell to accumulate water; TRI (terrain ruggedness index) quantifies local elevation difference as a proxy for terrain cost.

are discrimination metrics computed against whatever background is supplied to them, which is precisely the property this paper is about (Guillera-Arroita et al. 2015; Lobo, Jiménez-Valverde & Real 2008). The original pipeline set itself a target of spatial AUC > 0.75 under block cross-validation.

2.3 The experiment progression, as the object under examination

Seven iterative experiments (E007–E013) built the model whose evaluation we now examine. They are the object of the analysis, not its result, so we summarise them in one table and two paragraphs and move the per-iteration detail to the supplement. Table 3 states exactly which covariates each experiment used, read from the analysis scripts.

Two paragraphs of interpretation. Feature-only expansion plateaued (E007–E009): adding river distance helped (+0.036) and adding soil covariates hurt (−0.031). Background redesign then delivered the monotone rise (E010–E013) that we initially read as evidence that pseudo-absence realism dominates feature accumulation. The remainder of this paper asks whether that rise is real.

The hybrid design of E013 blends a TGB candidate pool with a regional quota and a

Exp.	elev	slope	twi	tri	aspect	river	road	Background design	Rep. spatial AUC
E007 v1	✓	✓	✓	✓	✓	–	–	uniform random, 1:5, 2 km buffer	0.659
E008 v2	✓	✓	✓	✓	✓	✓	–	uniform random, 1:5	0.695
E009 v3	✓	✓	✓	✓	✓	✓	–	uniform random, 1:5 (+clay/silt)	0.664 ↓
E010 v4	✓	✓	✓	✓	✓	✓	bg	TGB (major road)	0.711
E011 v5	✓	✓	✓	✓	✓	✓	bg	TGB, parameters swept	0.725
E012 v6	✓	✓	✓	✓	✓	✓	bg	TGB, expanded road classes	0.730
E013 v7	✓	✓	✓	✓	✓	✓	bg	hybrid = TGB pool + regional blend + hard negatives	0.768 best / 0.751 seed-avg

Table 3: Per-experiment covariate inclusion. ✓ = training feature; – = not used; *bg* = used to construct the background, never as a feature. Reproducibility: rasters in `data/processed/dem/`, presences in `data/processed/east_java_sites.geojson` (378 valid in bounds), deterministic 5-fold spatial blocking at 0.45°, pseudo-absence ratio 1:5 throughout. E010–E013 use `road_dist` to build the background while excluding it from the features.

fraction of 'hard negatives', sweeping `region_blend` $\in \{0.0, 0.3, 0.5, 0.7\}$ and `hard_frac` $\in \{0.0, 0.15, 0.30\}$. In this study, hard negatives are defined as points at a standardised environmental distance (z) of 2–5 from the mean site environment (Table 1); although the targeted proportion in the hybrid design is 30%, the realised proportion reached 0.62. This happens because the remaining 70% of points are drawn from a background pool in which about half of the members already have $z \geq 2$, not because of a particular bias of the TGB, whose proportion of such points (0.485) is close to that of the random background (0.503). This pool-composition effect bears on the absolute AUC values and is part of why a common evaluation background matters (Section 3.2).

2.4 The four roles of `road_dist`

`road_dist` is deliberately never a training feature, yet it does four jobs at once: it (i) defines the TGB acceptance probability, (ii) defines the hybrid background pool, (iii) defines the accessibility-proxy holdout split (E014, Section 2.5), and (iv) serves as a visibility proxy in the diagnostic suite. The original pipeline included diagnostic tests, named Test 1 and Test 3 in its code. In Test 1, the correlation between predicted suitability and the distances to volcanoes, to roads and to the nearest site is examined; the diagnostic is considered failed only if the absolute Spearman correlation exceeds 0.5. Test 3 compared model performance across road-distance quartiles; it was scored on the design's own background, so its result is subject to the artefact this paper describes and is not used here. The volcano-related component of Test 1 is reported in Section 3.8. Because the

variable used to build the background can also affect the reported metric values, the role of `road_dist` in building the background and in the diagnostics needs to be explained here, in the Methods, together with the other covariates. This is also what makes the approach specifically archaeological: the archaeological content is not in the label set, it is in the background design: a claim about where a survey could plausibly have detected a site had one been present.

2.5 The accessibility-proxy holdout (E014)

E014 is an *accessibility-proxy holdout*: the model is trained on sites ≤ 1 km from a road and tested on sites > 1 km away, with road accessibility standing in for discovery order (sites far from roads stand in for sites found late), so the split reflects differences in accessibility, not differences in time. The model yields an AUC of 0.755.

2.6 The refutation design: E217–E224

The eight experiments E217–E224 are the paper's centrepiece. E218 and E220–E224 were run from written designs with decision rules; only for E224 was the design committed to the public repository before the run (for the others, design and results were committed together), and E217 and E219 have no written design.

One deviation from the original pipeline must be stated before the results, because it bears on how they should be read. E217–E224 draw background points from a decimated raster lattice ($\sim$300 m spacing) rather than by the continuous-coordinate rejection sampling used in E007–E013. This was a deliberate choice: it gives every background design a single shared sampling frame, which is a precondition for the common-background comparison, and it keeps a factorial benchmark tractable. The cost is that absolute AUCs in E217–E224 are not directly comparable to the original E007–E013 values. The quantities we interpret are therefore *within-benchmark contrasts* (design against design, and own-background against common-background, under identical folds), not the absolute levels. Because the lattice is a departure from the original pipeline, E218 re-ran the comparison at half that spacing ($\sim$150 m, 5 seeds): the ladder persists on own backgrounds (random

0.711 → hybrid 0.757) and stays flat on the common background (random 0.705 → hybrid 0.704), so the conclusion does not depend on the lattice.

E217 – common-background benchmark. MaxEnt, XGBoost and RandomForest are run under *identical* spatial-block folds and *identical* background designs as E007–E013, but each design is scored against a *common* evaluation background shared across designs, so that the comparison is identified. A second feature set (`terrain` alone, 5 vars vs `terrain_river`, 6 vars) lets us compare a feature effect against a background-design effect.

E218 – the evaluation artefact, isolated. Three training designs are each scored against four evaluation backgrounds (own + three foreign), exposing how much of each design's apparent lead comes from being scored on its own background.

E220 – does the selection criterion have an interior optimum? We vary the share of hard negatives in the background from 0 to 1 and ask whether the AUC-on-own-background criterion ever tells a researcher to stop before the end of that range.

E221 – seed stability of the priority map. Across random seeds, how much of the top-decile priority map turns over, and how much does averaging seeds reduce it?

E222 – synthetic ground truth. Four synthetic worlds of known intensity surface let us compare *reported* AUC (own-background) against *true* recovery (common-background) where ground truth is not in question.

E223 – confidence intervals and regularisation. Bootstrap and equivalence CIs bound the ladder gain; a MaxEnt regularisation sweep ($\beta = 0.5$–4.0) tests whether the artefact is an overfit of an unregularised model.

E224 – testing one explanation of the TGB null (Section 3.5). We proposed that the TGB null in E222 was structural, tested it, and the test failed; the null is reported as unexplained.

3 Results

3.1 The reported ladder

Reproducing the claim under examination: Table 4 and Figure 2 show the rise from E007 (0.659) to E013 (seed-averaged 0.751; original run 0.768, highest of 20 further seeds 0.774; TSS 0.507 ± 0.167), with 20-seed stability in the range 0.729–0.774. Even the worst seed exceeded a spatial-interpolation null (DKNS, 0.646) by +0.083, and the model beat the three null models in Table 5. Taken at face value, this sequence supports the reading that more realistic background designs improved the model.

Exp.	Pseudo-absence strategy	XGB AUC	RF AUC	Best AUC	XGB TSS	RF TSS	Status
E007	Random	0.659 ± 0.077	0.656 ± 0.090	0.659	0.318 ± 0.126	0.314 ± 0.133	REVISIT
E008	Random + river feature	0.685 ± 0.074	0.695 ± 0.107	0.695	0.345 ± 0.135	0.379 ± 0.200	REVISIT
E009	Random + soil features	0.664 ± 0.049	0.643 ± 0.054	0.664	0.337 ± 0.083	0.312 ± 0.072	REVISIT
E010	TGB (major road)	0.711 ± 0.085	0.699 ± 0.081	0.711	0.384 ± 0.150	0.380 ± 0.130	REVISIT
E011	TGB tuned	0.725 ± 0.084	0.716 ± 0.081	0.725	0.447 ± 0.184	0.408 ± 0.147	REVISIT
E012	TGB tuned (expanded)	0.730 ± 0.085	0.724 ± 0.081	0.730	0.420 ± 0.170	0.413 ± 0.152	REVISIT
E013	Hybrid bias-corrected	0.768 ± 0.069	0.742 ± 0.070	0.768	0.507 ± 0.167	0.458 ± 0.126	SUCCESS

Table 4: Experiment progression E007–E013 (the ladder under examination).

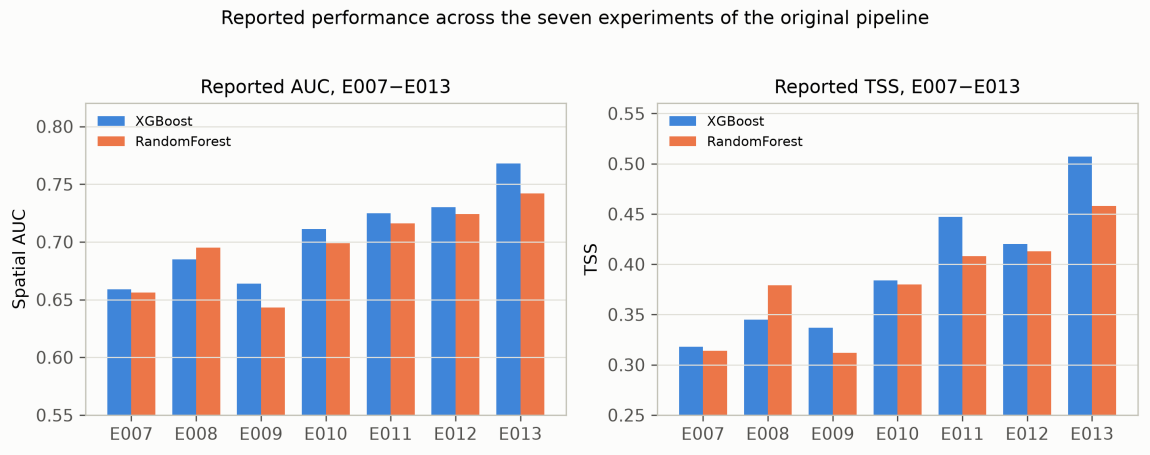


Figure 2: Reported AUC and TSS across the seven experiments of the original pipeline (E007–E013; Table 4). This is the ladder under examination; Section 3.2 asks whether the rise *between* E007 and E013 is real.

3.2 The ladder is an evaluation artefact

The part of the rise attributed to background redesign does not survive a common evaluation background. On real data, redesigning the background while holding a common

Model	AUC	Std	Gap vs E013
Random (chance)	0.500	0.000	+0.251
Heuristic (river < 2 km)	0.581	0.054	+0.170
DKNS (spatial interpolation)	0.646	0.032	+0.105
E013 XGB (seed-avg)	**0.751**	0.013	—
E013 XGB (best run)	0.768	0.069	—

Table 5: Null-model comparison under spatial CV. Gaps are computed against the E013 seed-average (0.751), not the best run, because a single run is not a reportable quantity (Section 3.7). The nulls are scored on a regular grid of all cells, while the E013 column is the model's own-background seed-average — so the margins shown still mix evaluation backgrounds, a defect corrected in Section 3.2, where the consequence for this level comparison is stated. DKNS (distance to known nearest site) uses site locations as an unfair upper bound.

evaluation background fixed contributes −0.0142 **AUC** averaged over the three algorithms, whereas adding the river feature contributes +0.0424 **AUC**, positive in 12/12 paired comparisons (the redesign contribution is +0.0054 on the terrain-only feature set and −0.0142 on the full set; we state the feature set so a reader does not encounter the difference unannounced). The hybrid design beats the random design in 3/3 algorithms when each is scored on the *hybrid* evaluation background, and in **0/3** on each of the uniform, target-group and stratified evaluation backgrounds. The advantage disappears as soon as a design is scored on any background other than its own.

How large is own-background inflation? For the hybrid design, the last step of the ladder, the inflation from own-background evaluation is positive in **60/60** seed×algorithm cells (mean +0.037, range +0.005 . . . + 0.084), the same order of magnitude as the entire reported E007→E013 gain. This own-background inflation is specific to that design, and we state the scope rather than let it read as general: the TGB design scored *higher* on the uniform background than on its own (mean −0.0054, 22/60 cells positive). The TGB steps of the ladder (E010–E012) are not explained by own-background inflation either: in the benchmark TGB scores almost exactly as random does even on its own background (+0.003, E217), so their original gain is not reproduced rather than explained. In the synthetic worlds, a model of the hybrid design scored on its own background is reported higher than its truth in **all 240 runs (median** +0.238**)**. For the random and TGB designs the gap is smaller and not always positive (52/60 and 51/60

runs; medians +0.020 and +0.014), and part of it comes from the difference between the observed and the true presence samples rather than from the background. Over all 360 runs of the two synthetic surfaces the count is 343/360 (Figure 3). We reject the reported ladder gain (+0.092) directly: equivalence confidence intervals exclude it in **12/12** algorithm×evaluation-background cells, and a bootstrap that resamples presence *blocks* rather than points (30 replicates, of which one was skipped because fewer than 50 out-of-bag sites remained, so 29 were used; fitted in-bag and scored out-of-bag) rejects it for all three algorithms, with CI upper bounds of +0.008, +0.025 and +0.026. Twenty-nine replicates is a small bootstrap; we report it because the rejection margin is wide relative to the resampling noise, not because the interval endpoints are precisely estimated. Changing the MaxEnt regularisation β across 0.5–4.0 changes nothing (-0.020 to -0.022, 1/10 runs positive), so the artefact is not an overfit of an unregularised model.

One consequence for the level claim reported in Table 5 must be stated plainly. That table scored the E013 model on its own (hybrid) background but scored the null models on a regular grid of all cells; given the inflation above, the +0.105 seed-average margin over the DKNS null mixed the two backgrounds. Held to a common (uniform) background, the sampling on which the nulls were scored, the E013 design family reaches 0.706 (XGBoost, the primary learner, seed-mean; E218), which still exceeds the spatial-interpolation null (0.646), but the margin shrinks from +0.105 to roughly +0.06, and the reported worst-seed margin (+0.083) moves toward the +0.03 detection floor of Section 3.6. The two figures are not measured under identical sampling (0.751 comes from the original pipeline and 0.706 from the lattice-based benchmark, Section 2.6), so the shrinkage should be read as an order of magnitude, not a difference of differences. The qualitative level claim, that the environmental features capture structure beyond nearest-site interpolation, survives the correction; its size does not. We therefore treat Table 5 as an upper bound on the model's advantage over the nulls, not a robust floor.

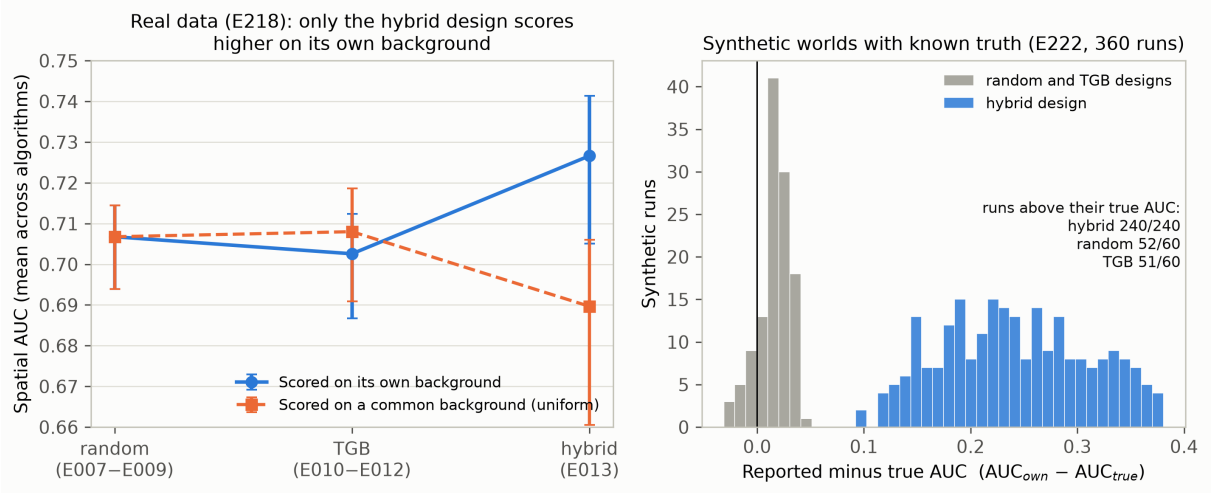


Figure 3: The evaluation artefact. Left: reported AUC (evaluated on a design's own background) against the same designs held to a common background (uniform). Only the hybrid design scores higher on its own background. Right: reported minus true AUC in 360 synthetic runs of known ground truth, by design (hybrid 240/240 runs positive; random 52/60; TGB 51/60).

3.3 The selection criterion has no interior optimum

Why does the ladder exist at all? The selection criterion (AUC computed on a design's own background) has no interior optimum: its maximum always sits at the end of whatever range is swept, in both the synthetic and real sweeps (Figure 4). In the synthetic worlds the criterion rises monotonically with the hard-negative share (0.7367 → 0.8904); on the real data it is monotone from 0.1 upward, with one dip of −0.0071 at 0.0 → 0.1. Because the criterion never points inward, it never tells a researcher to stop.

The practical consequence is mild at the largest share tried in E013 (0.30). E013 stopped at `hard_frac` = 0.30 because its grid stopped there, not because the criterion said to; the cost of that stop, against the configuration with the highest true AUC on the same grid, is small (+0.0044 real; median +0.0000 synthetic, mean +0.0012, maximum +0.0088), and on that grid the criterion does not pick the truth-best design in 29/60 cases but the differences are essentially zero. The danger is structural: if the hard-negative share were pushed further, the cost is large (+0.0973 real; +0.1937 synthetic median, 100% positive), and the criterion provides no brake. We note what the criterion does *not* do. In the synthetic worlds, the design it selects (hybrid(1.0), the hybrid design with a hard-negative share of 1.0) never has the lowest true AUC of the designs compared (0/60

cases); the lowest is hybrid(0.0) in 50/60, hybrid(0.3) in 8 and hybrid(0.7) in 2 of 60 cases. The selected design costs +0.194 **$\mathbf{AUC_{true}}$ against the best available**, but it is not the worst.

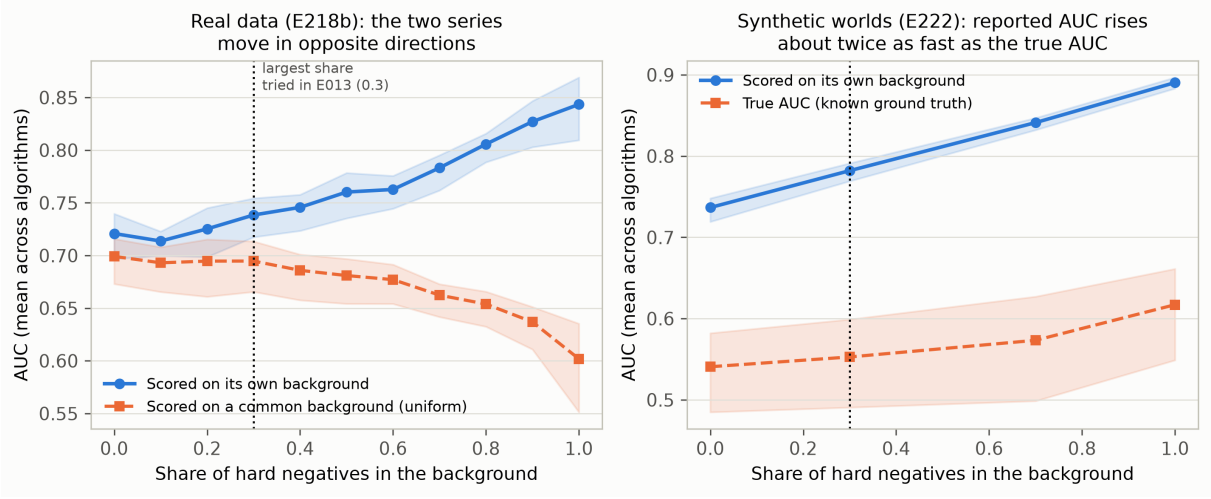


Figure 4: The selection criterion on a design's own background (solid) keeps rising to the end of the hard-negative range; the comparison series (dashed) falls on real data and rises only about half as fast on synthetic data, so the criterion has no interior optimum. Left: real data, scored on a common background; the two series move in opposite directions. Right: synthetic ground truth; reported performance moves about twice as fast as true performance. The dotted vertical line marks 0.30, the largest share tried in E013; the range tried here ends at 1.0.

3.4 Reported and true performance diverge

In the synthetic worlds the reported number moves about **twice** as fast as the truth as the hard-negative share rises (+0.1538 reported vs +0.0764 truth; endpoint ratio 2.01×, per-run OLS slope 2.12×, per-run median ratio 2.00×; the three estimators bracket 'about twice'). On the real data the two move in **opposite directions** (+0.1227 reported vs −0.0973 common-background), which is a stronger result. A pre-registered prediction that own-background inflation rises with the hard-negative fraction in every regime **failed** (pooled Spearman 0.4395 $<$ 0.5) and is reported as failed.

3.5 The target-group-background null, reported as unexplained

Our synthetic worlds gave TGB exactly the condition its theory requires (a known, survey-biased observation process), and TGB did not help. Over the synthetic runs, TGB minus random on top-decile map agreement (Jaccard) is -0.010, positive in 46.7% of paired comparisons. In regional-bias regimes TGB is neutral rather than harmful (true AUC, TGB minus random: World C -0.0010, 56.7% positive; World D $+0.0022$, 73.3% positive), and it wins only under model misspecification (Jaccard 0.4504 vs 0.4458 in World B), which we report because it is true but do not build on.

One possible explanation was that, because `road_dist` is not a model feature, the survey-bias factor is unrepresentable in feature space, so TGB has nothing to cancel. We pre-registered a test with both decision branches written first, then ran it (E224): the same synthetic worlds with `road_dist` added to the features. It made no difference. TGB minus random was -0.0217 with `road_dist` in the features (30% positive) and -0.0254 without (30% positive). We therefore report the null as **unexplained**, and we disclose the limitation of our own test: `road_dist` correlates $+0.49$ with `river_dist` (and $+0.31$ with elevation), so the model already had partial access to the bias signal in the control condition; a clean test needs a survey-effort surface orthogonal to terrain by construction, which is future work. We resisted re-interpreting the secondary metric (`auc_true`, which did rise from 36.7% to 60% positive) because the pre-registered primary metric was map Jaccard, and it did not support that reading.

3.6 Confidence intervals and a detection floor

The equivalence CIs that reject the $+0.092$ gain (Section 3.2) also bound what we can rule out: at $n = 378$ the spatially blocked bootstrap cannot exclude effects below about $+0.03$. Findings smaller than that are indeterminate at this sample size, not absent.

3.7 Priority maps change with the seed; averaging seeds reduces this

A model whose numbers are artefactual still produces a priority map, and in heritage practice the map is what gets used. Across random seeds, 28.1%–47.4% of top-decile cells turn over between seed pairs, so a single-seed map is not a sound basis for a survey decision. The remedy is cheap and we now make it a protocol: seed-ensemble. Measured against the 10-seed ensemble, which contains the seeds being averaged, the overlap passes 0.90 after 4–7 seeds, but that comparison shares seeds with its reference and so flatters the result. Two ensembles that share no seed agree less: Jaccard 0.55–0.70 for single seeds, and 0.76 (XGBoost), 0.85 (MaxEnt) and 0.86 (RandomForest) for five seeds each (Figure 6; E236). Averaging clearly helps, so we recommend at least seven seeds as a floor, together with reporting the agreement between ensembles that share no seed, rather than assuming that a given number of seeds makes the map stable. This ensemble reveals a useful distinction between a *robust core* (the consistent core area) and a *contingent fringe* (the peripheral area that depends on the background design). The robust core is defined as the cells that fall in the top 10% under all three background designs (random, TGB and hybrid), each map being the average of 10 seeds; the contingent fringe is the cells that fall in the top 10% under exactly one design. Observed site density is higher in the core than in the fringe: 1.93× for RandomForest (30.7 vs 15.9 sites per 1,000 km^2), 4.34× for XGBoost (40.8 vs 9.4) and 5.62× for MaxEnt (31.7 vs 5.65; Figure 5). This difference indicates spatial consistency, not independent validation, because the sites used to compute these densities were also used to train the models.

3.8 Corrections to two volcanic diagnostics

Two diagnostics of the original pipeline are corrected here.

The volcano inventory. The analysis code hard-coded 7 volcanic centres; the canonical inventory contains **13** inside the paper's own 111–115°E bounds (adding Lawu, Wilis, Kawi-Butak, Penanggungan, Iyang-Argapura, Baluran), and two of the additions sit inside the Malang–Mojokerto site concentration. `volcano_dist` is not a training feature,

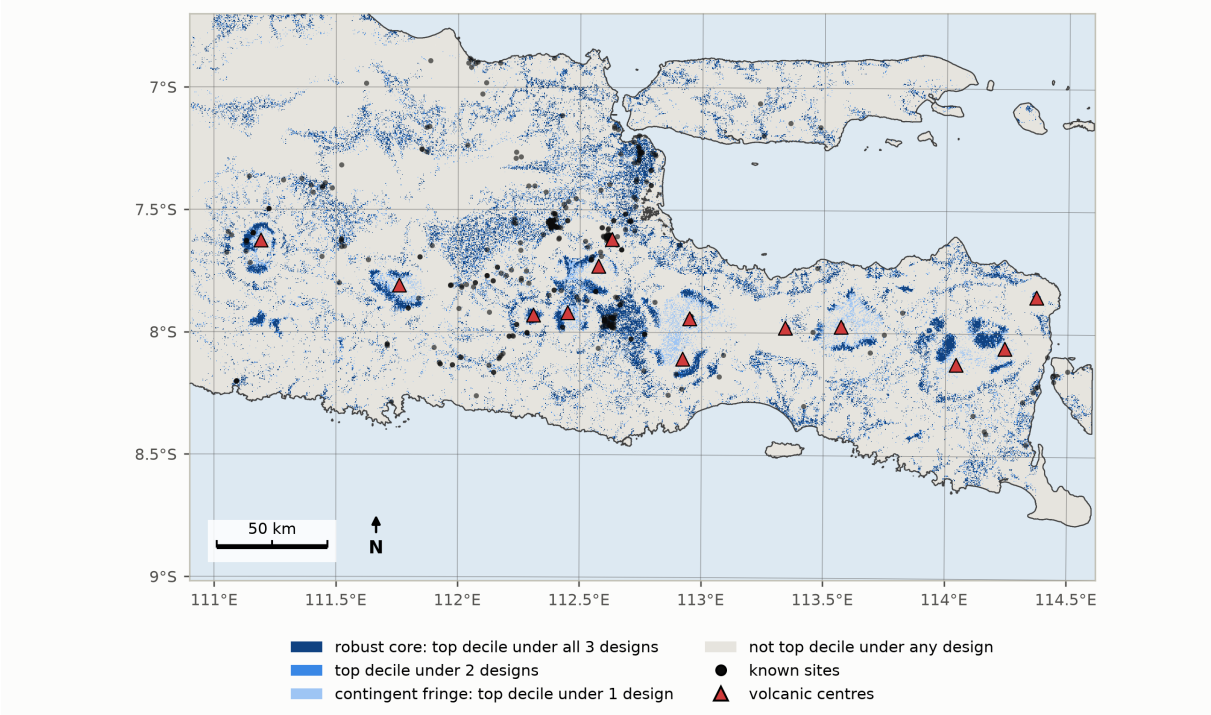


Figure 5: Robust core and contingent fringe for XGBoost under a 10-seed ensemble, cells coloured by how many of the three background designs placed them in the top decile. The core concentrates on the accessible lowlands that hold most of the known records; the contingent fringe, where the background designs disagree, lies higher (median elevation 491 m for XGBoost and 1,107 m for MaxEnt).

so model AUCs are unaffected; the Test 1 tautology diagnostic is. Recomputed on the canonical inventory, the suitability–volcano-distance Spearman correlation is $\rho = -0.281$ (legacy 7-volcano re-run: -0.243). The verdict is unchanged: $|\rho| < 0.5$, so the test still passes.

A correlation that is not consistent across seeds. A single model instance gives a Spearman correlation of -0.163 for the same test. However, repeating the analysis with five seeds on the same seven-volcano inventory gives -0.243 (E219). The re-run also uses the lattice-based background sampling of Section 2.6, so the difference may come from the seeds or from the change of sampling; the two cannot be separated here. Either way, a single model instance is not a sound basis for the diagnostic. The corrected values reported here are seed-ensembled; the E007–E013 ladder in Table 4 is the original pipeline, reproduced as the object under examination.

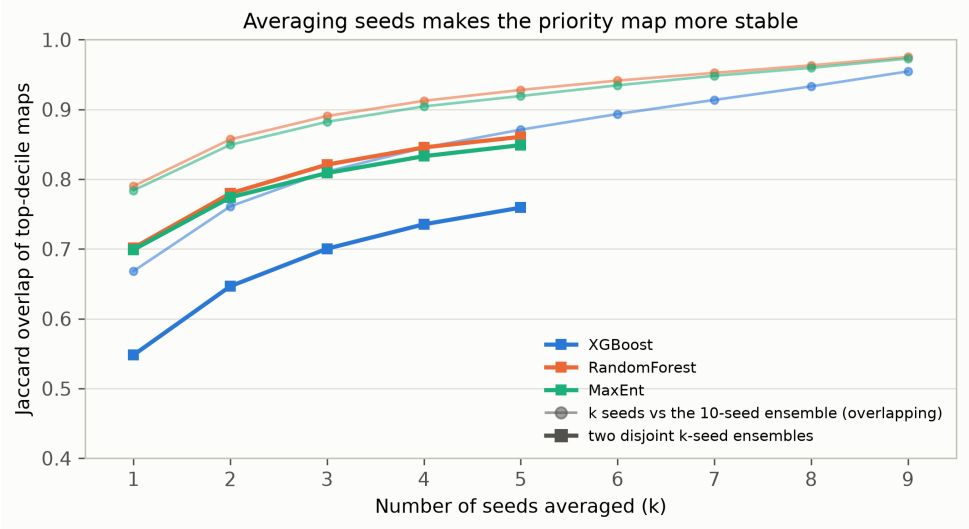


Figure 6: Agreement of top-decile priority maps as seeds are averaged, one colour per learner. Thin lines: an ensemble of k seeds against the full 10-seed ensemble, which contains those seeds (E221). Thick lines: two ensembles of k seeds that share no seed (E236; at most five seeds each from ten). The thick lines are the fair measure of stability.

3.9 Volcanic setting or elevation?

This analysis tests whether the observed suitability pattern reflects a volcanic setting or simply the effect of elevation. Distance to a volcanic centre (`volcano_dist`) is not used as a training feature, so any pattern in the model's predictions that relates to a volcanic setting must come through the terrain variables the model uses. Near land is defined as land within 20 km of a Quaternary volcanic centre, and far land as land 40 km or more from one; the analysis includes the 30 canonical volcanic centres, including those outside the map frame. The comparison is made within each 200 m elevation band, using coarsened exact matching on slope, the terrain ruggedness index (TRI) and the topographic wetness index (TWI). The analysis uses the stored 10-seed ensemble maps, without retraining any model. Uncertainty is estimated with a spatial block bootstrap of 1,000 replicates; the decision threshold of 0.05 was set before the analysis, and the design was pre-registered in the public repository. The verdict labels follow the pre-registered rule: SPECIFIC if the 95% CI excludes zero and the size of the difference is at least 0.05; NOT SPECIFIC if the whole 95% CI lies between -0.05 and $+0.05$; and INCONCLUSIVE for any result that meets neither criterion.

Above 1,000 m the study area contains about 3,435 km^2 of land, of which 97.5% lies

within 20 km of a Quaternary volcanic centre and only 0.4% lies 40 km or more from one (Figure 7A). Land far from Quaternary volcanic centres covers only about 15 km^2 at 1,000–1,200 m and none above 1,200 m. This means that, in this study area, the effects of elevation and of proximity to a Quaternary volcanic centre cannot be adequately separated above about 1,000 m. This study therefore does not attribute the pattern there specifically to either elevation or volcanic setting. This limit of separability can shift between 800 and 1,200 m, depending on the minimum-area criterion used.

Below 1,000 m, the difference in suitability between land near and far from Quaternary volcanic centres is +0.049 for the random design (95% CI +0.024 to +0.077), +0.049 for TGB and +0.060 for hybrid (95% CI +0.030 to +0.091; Figure 7B). Under the pre-registered rule, the hybrid result meets the criterion for SPECIFIC, but the summary across designs remains INCONCLUSIVE, because two of the three designs do not give a firm conclusion under that criterion. The two pre-registered sensitivity definitions give smaller differences: −0.002 (95% CI −0.047 to +0.046; NOT SPECIFIC) with near land within 10 km and far land beyond 30 km, and +0.016 (INCONCLUSIVE) with 15 and 30 km. Post hoc variants give +0.005 to +0.07, depending on the zone definitions and the matching procedure; when distance to rivers is also matched, the difference falls to +0.028. No direction is therefore claimed below 1,000 m. In the uplands (400–1,000 m), the far comparison land lies mostly in the Southern Mountains, an old volcanic arc of Eocene–Miocene age with karst landforms (Smyth, Hall & Nichols 2008), so it would not be correct to call it non-volcanic. In addition, the contingent fringe of the MaxEnt maps (Section 3.7) has a median elevation of 1,107 m, above the limit of reliable separation; that pattern therefore cannot be interpreted specifically as either an elevation effect or a volcanic effect. The limited power of the test also does not allow a firm conclusion about whether the disagreement between designs differs between near and far land.

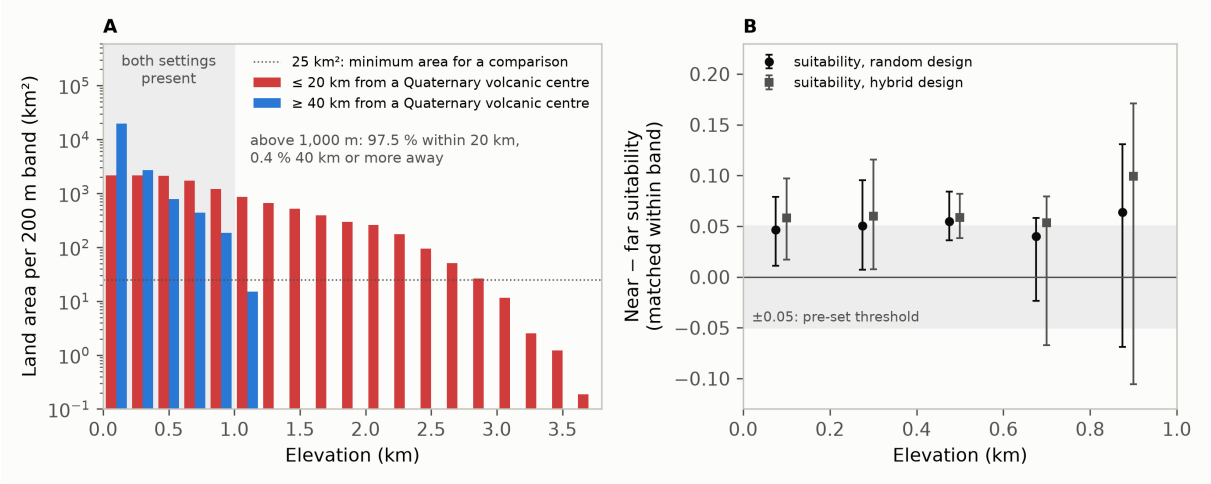


Figure 7: (A) Land area in each 200 m elevation band for land near ($\leq$ 20 km) and far ($\geq$ 40 km) from a Quaternary volcanic centre. (B) Difference in suitability between near and far land after matching on elevation, slope, TRI and TWI, with 95% confidence intervals (XGBoost); the grey band marks the decision threshold of ± 0.05.

3.10 What a suitability surface cannot show, and a quota design that failed

A suitability surface cannot distinguish the mechanisms that cause a site to go unobserved (Section 1.2). Low predicted suitability near a volcano therefore cannot, by itself, be read as evidence of buried settlement. We also report, without softening, a quota background design we tested in the regime it was built for (regional survey bias) and in its most favourable regime (balanced-regional truth): it failed in both (World C: -0.2457 AUC_{true}, -0.4688 Jaccard, 0/30; World D: -0.2027 AUC_{true}, -0.2826 Jaccard, 0/30).

4 Discussion

4.1 Hold the evaluation background fixed

The central methodological finding is that a background-design comparison is not identified unless the evaluation background is held fixed across the designs being compared, and that the artefact this produces is large enough to masquerade as a result. We are not the first to observe that background choice drives reported performance (Hijmans 2012; Lobo, Jiménez-Valverde & Real 2008; Phillips et al. 2009); what we contribute is the mechanism

(a selection criterion with no interior optimum, Section 3.3) and a direct quantification of the inflation it produces (for the hybrid design, all 240 synthetic runs and 60/60 real seed×algorithm cells). The practical recommendation follows directly: *report background designs on a common evaluation background, and state the evaluation availability domain of every metric*. This is the accessible-area requirement of Barve et al. (2011) applied to evaluation rather than to fitting: the domain over which a metric is computed is part of the metric, and a number reported without it cannot be compared with another. We learned this from our own model.

4.2 The archaeological content is in the background

What makes a presence–background model archaeological, rather than a generic classifier run on coordinates, is not the label set. It is the background design: a statement about where a survey could plausibly have detected a site had one been present. That is what TGB and hybrid designs encode, and it is why they are archaeological choices rather than sampling conveniences. The sharp corollary, which this paper exists to state, is that because the archaeological content lives in the background, and because the model was evaluated against that same background, the very thing that made the design archaeological is what contaminated its evaluation.

4.3 A suitability surface is not a discovery map

The output is a suitability surface with declared limits, not a map of where sites are. Low suitability does not mean 'buried' or 'destroyed'; high suitability does not mean 'a site is there' (Section 1.2).

4.4 Implications for heritage management in East Java

Under Indonesia's Cagar Budaya framework (UU 11/2010), surveys by the regional heritage preservation office (Balai Pelestarian Cagar Budaya, BPCB) inform permitting and rescue prioritisation. A suitability surface could in principle support that workflow, but only under two conditions this study makes explicit. First, because priority maps turn

over by 28–47% between seeds (Section 3.7), a single-seed map is not a defensible basis for a permit; the seed-ensemble protocol (at least seven seeds, with the agreement between disjoint ensembles reported) is a precondition for operational use. Second, because the model reports suitability and not presence, the robust core (1.9–5.6× the site density of the fringe) is a prioritisation aid, not a prediction: it ranks where to look, not what will be found.

4.5 The corrected protocol

We summarise the protocol that follows from the above, which is the constructive contribution of this study:

1. **Hold the evaluation background fixed** when comparing background designs; report own-background numbers only alongside common-background numbers.
2. **Declare the availability domain** of every metric (background design, benchmark, selection rule).
3. **Seed-ensemble priority maps** (at least seven seeds) and report the agreement between ensembles that share no seed; report a robust core and contingent fringe, never a single-seed map.
4. **Do not let the selection criterion decide where to stop**: because it has no interior optimum, justify where to stop varying a background-design parameter on external grounds, not by the criterion's own maximum.

4.6 Limitations

1. **Calibration gap between synthetic and real.** At a hard-negative share of 0.30 (the largest tried in E013) the selection criterion picks hybrid on real data but random on synthetic data, so the synthetic worlds do not reproduce the real-data selection behaviour there. This limits transferring the E222 conclusions back to the real case.

2. **The truth-slope sign is regime-contingent**: negative on real data, positive on synthetic. We do not claim 'hard negatives are always bad.'

3. **Four synthetic regimes are not a universal result.** Across the four worlds (road-bias, misspecified, regional-bias, balanced-regional-truth; $n \approx 300$–500), no background design exceeded uniform on truth recovery by a meaningful margin (aggregate difference ≈ -0.0004 AUC; in the two road-bias and misspecified surfaces TGB scored above random in 27/60 runs). This is not a refutation of target-group background in general (Phillips et al. 2009); it is a result for the regimes and sample sizes tested.

4. **A detection floor of** $\sim +0.03$ at $n = 378$: smaller effects are indeterminate, not absent (Section 3.6).

5. **The Boyce index is a direction check, not a selector.** It tracks the truth only loosely (median rank correlation +0.50 with map agreement and +0.54 with true AUC, E222), so we use it only to confirm direction.

6. Above about 1,000 m (the limit shifts between 800 and 1,200 m, depending on the minimum area used), **elevation and proximity to a Quaternary volcanic centre cannot be adequately separated** in the study area; the model's prediction pattern there, including the high-elevation fringe in Figure 5, therefore cannot be attributed specifically to either factor (Section 3.9).

7. **The two-stage 'suitable but absent' decomposition** is declared future work: it tests *why* sites are absent, which requires site-level prediction, which this design does not support, and burial-depth data at survey resolution are not yet available.

8. **External transfer beyond East Java is untested.**

5 Conclusions

In this study, the reported AUC of a presence–background model for East Java rose overall from 0.659 (E007) to 0.751 (E013, mean over seeds) across seven experiments, and the

rise was attributed to increasingly realistic background designs. When all designs are evaluated on a common background, changing the background design contributes −0.014 AUC and adding distance to rivers contributes +0.042. On a common background the change of design adds nothing. The last design, the hybrid, scores higher only when it is evaluated on its own background points, which is an evaluation artefact, and the gains of the earlier target-group designs are not reproduced. The criterion used to choose a design, AUC on the design's own background, has no optimum inside the range tested, so it gives no indication of when to stop adding hard negatives. A suitability surface also cannot distinguish the reasons a site goes unobserved, so low predicted suitability near volcanoes is not evidence of buried settlement; and above about 1,000 m, elevation and proximity to a Quaternary volcanic centre cannot be adequately separated in East Java.

The contribution of this study is therefore a protocol, not a new prediction of site locations. Background designs should be compared on a fixed evaluation background, the availability domain of every metric should be stated, and priority maps should be averaged over at least seven seeds, with the agreement between ensembles that share no seed reported. We applied these checks to our own earlier analyses, where they changed the conclusion; they need no new data and can be applied to other presence–background comparisons.

Supplementary Materials

Supplied as a single document, `supplementary_tables_v0.3.pdf`. Every table in it is generated directly from the raw result files by `build_supplement.py` and `build_e235_tables.py` (included in the code repository); no value is retyped.

- **Table S1:** Seed stability — XGBoost and RandomForest AUC and TSS across the 20 alternate seeds for E013.
- **Table S2:** Block-size sensitivity — spatial CV at ∼40, ∼50 and ∼60 km, with 95% intervals.

- **Table S3:** E217 common-background benchmark — every design × feature set × algorithm, scored on its own background and on a common background.
- **Table S4:** E218 own- versus common-background matrix — three training designs against four evaluation backgrounds.
- **Table S5:** E222 synthetic worlds — reported against true AUC by background configuration and algorithm.
- **Table S6:** Per-experiment design matrix for E217–E224 (extends Table 3), including which experiments used `road_dist` as a feature.
- **Table S7:** E235 elevation bands — land near and far from Quaternary volcanic centres in each 200 m band, and the matched near − far differences (Section 3.9).
- **Table S8:** E235 pooled differences — by zone definition and algorithm, with the pre-registered verdicts.

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Competing Interests

The authors have no competing interests to declare.

Authors' Contributions

M.A. conceived the study and its design, directed and checked the analyses (implemented with AI assistance, as declared below), curated the data, and prepared the manuscript.

G.F.G. contributed to methodology refinement, code review, and manuscript editing. Both authors read and approved the final manuscript.

Data Availability

Archaeological site data: the compiled East Java site dataset is in the repository at `data/processed/east_java_sites.geojson` (378 with valid covariates in bounds). Sources: OpenStreetMap (ODbL), Wikidata (CC BY-SA), Wikipedia Indonesia, BPCB Jawa Timur publications.

Environmental covariates: Copernicus GLO-30 DEM and derived layers from the Copernicus Programme (ESA), `https://dataspace.copernicus.eu/`. OpenStreetMap road and river data, `https://openstreetmap.org` (ODbL).

Code: all source code, including E217–E224, E235 and the verification script, is at `https://github.com/neimasilk/volcarch-repo` (MIT licence).

The G1 verification script is `verify_headline_numbers.py`.

Python dependencies are pinned in `requirements_submission_lock.txt`.

Interactive dashboard: `https://volcarch-dashboard.streamlit.app`.

Declaration of AI Use

We used Claude (Anthropic) throughout this study. It wrote and debugged the analysis code, produced the figures, drafted the designs of the experiments described in Section 2.6, checked the references against Crossref and the cited papers, and re-derived every reported number from the raw result files. It also helped with the text. The earlier drafts of the Methods, Results and Discussion were generated with Claude from our analyses and notes, and were then revised by us. In this version, new passages were written by the first author in Indonesian and translated into English with Claude; the abstract, the conclusions and a number of corrections were drafted with Claude at the first author's request. The research questions, the decisions about what to test and what to claim, and the interpretation are ours. We checked every statement and number against the results and take full responsibility for the content.

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Figure Alt Text

- Figure 1: Relief map of East Java with sites and volcanoes.
- Figure 2: Bar charts of AUC and TSS across seven experiments.
- Figure 3: Two panels comparing own-background and common-background model scores.
- Figure 4: Reported versus true performance as hard negatives increase.
- Figure 5: Map of East Java coloured by agreement between designs.
- Figure 6: Map agreement rising with the number of seeds averaged.
- Figure 7: Land area by elevation band and matched suitability differences.